Now, there is a feature of these slingshots that could be very unpleasant. Indeed, deadly: tidal forces (Chapter 4).

To change velocities by as much as $c/3$ or $c/4$, the Ranger must come close enough to the small black hole and neutron star to feel their intense gravity. At those close distances, if the deflector is a neutron star or is a black hole with radius less than 10,000 kilometers, the humans and the Ranger will be torn apart by tidal forces (Chapter 4). For the Ranger and humans to survive, the deflector must be a black hole at least 10,000 kilometers in size (about the size of the Earth).

Now, black holes that size *do* occur in Nature. They are called intermediate-mass black holes, or IMBHs, and despite their big size, they are tiny compared to Gargantua: ten thousand times smaller.

So Christopher Nolan should have used an Earth-sized IMBH to slow down the Ranger, not a neutron star. I discussed this with Chris early in his rewrites of Jonah's screenplay. After our discussion, Chris chose the neutron star. Why? Because he didn't want to confuse his mass audience by having more than one black hole in the movie. One black hole, one wormhole, and also a neutron star, along with *Interstellar*'s other rich science, all to be absorbed in a fast-paced two-hour film; that was all Chris thought he could get away with. Recognizing that strong gravitational slingshots *are* needed to navigate near Gargantua, Chris included one slingshot in Cooper's dialog, at the price of using a scientifically implausible deflector: the neutron star instead of a black hole.

Intermediate-Mass Black Holes in Galactic Nuclei

A 10,000-kilometer IMBH weighs about 10,000 solar masses. That's ten thousand times less than Gargantua, but a thousand times heavier than typical black holes. These are the deflectors Cooper needs.